Title: Road Accident Severity Prediction

(Domain - AIML + MERN)

1. Project Overview

Title: *Road Accident Severity Prediction using AI/ML & MERN Stack*

Objective:
To predict the severity level of a road accident (e.g., *Minor, Serious, Fatal*) based on various factors such as:

- **Road conditions** (wet, dry, icy, under construction, etc.)
- **Weather conditions** (rainy, sunny, foggy, snowy, etc.)
- **Vehicle information** (type, age, speed at the time, safety features)
- **Time factors** (day/night, rush hour, weekend, etc.)
- **Location data** (urban/rural, highway, city center)
- **Traffic data** (traffic density, signals, lane numbers)

The system will use historical accident datasets to train a **Machine Learning model**.
The prediction can be used by **smart traffic systems, government agencies, and insurance companies** to **reduce accident impacts** and **plan preventive measures**.

2. What Will the Project Do?

- **Input:** User or system feeds details of road, weather, vehicle, and time factors.
- **Processing:**
    - The ML model (trained on past accident data) processes the inputs.
    - Predicts the **accident severity category**:
        1. *Minor Injury*
        2. *Serious Injury*
        3. *Fatal Accident*
- **Output:**
    - Displays the predicted severity.
    - (Optional) Suggests safety measures (e.g., "Reduce speed due to icy roads").
- **Frontend (MERN):** User-friendly web dashboard where:
    - Authorities can enter accident-related info.
    - View predictions & analytics graphs.
- **Backend (Node.js + Express):**
    - Handles API calls between frontend and ML model.
- **ML Model (Python / scikit-learn / TensorFlow):**
    - Trained on dataset to make predictions.

3. High-Level Working (For College Review)

You can present this *without* implementation details in **three simple layers**:

a) Data Collection Layer

- Public datasets (Kaggle, government accident reports)
- Fields like weather, road condition, vehicle type, etc.

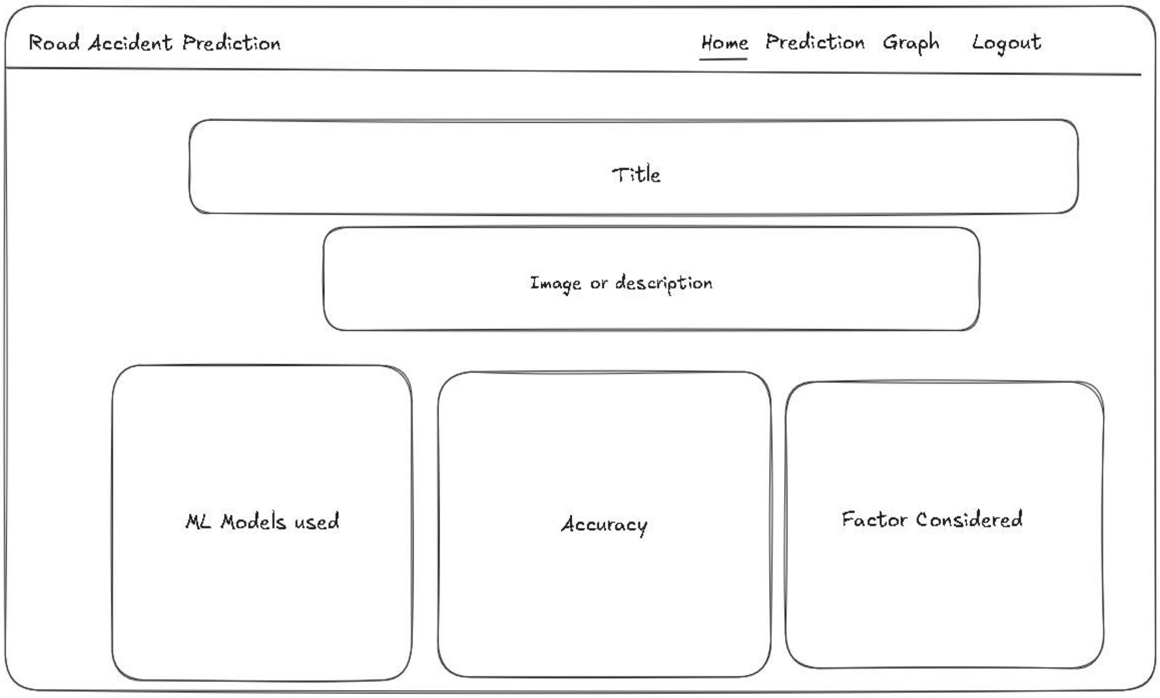
b) Prediction Layer (AI/ML)

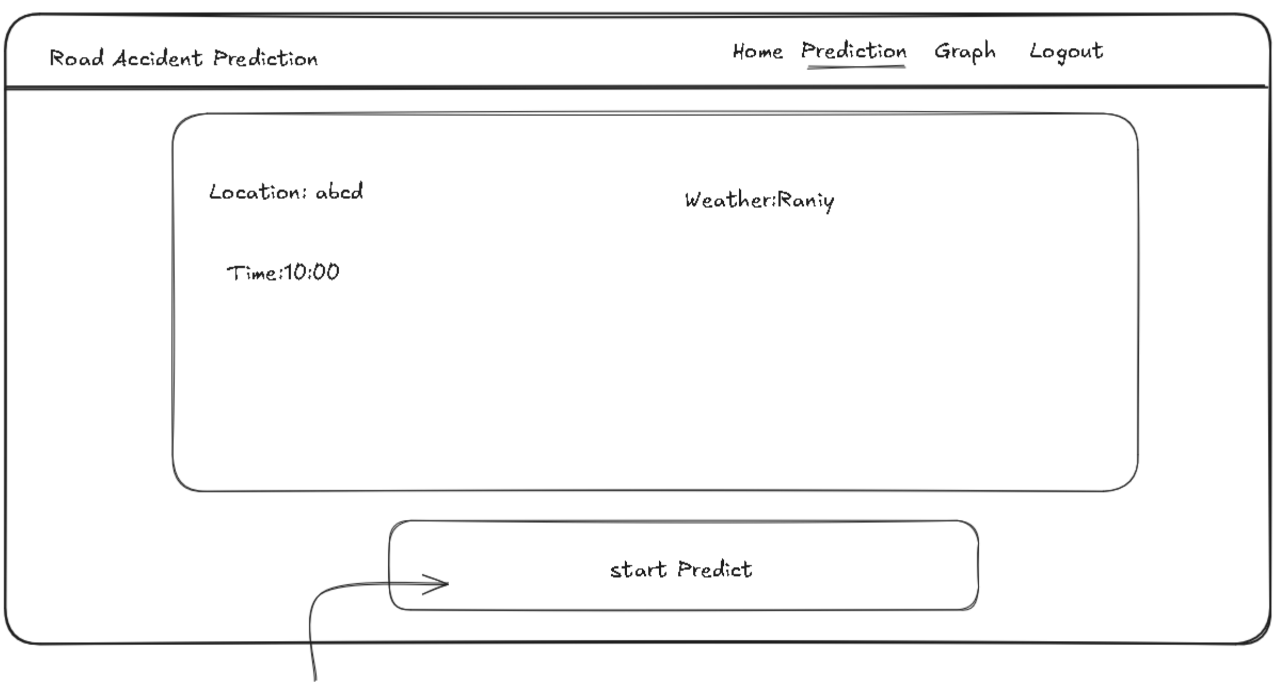
- Train a classification model (e.g., Random Forest, XGBoost, Neural Network)
- Input: New accident data → Output: Severity category

c) Application Layer (MERN)

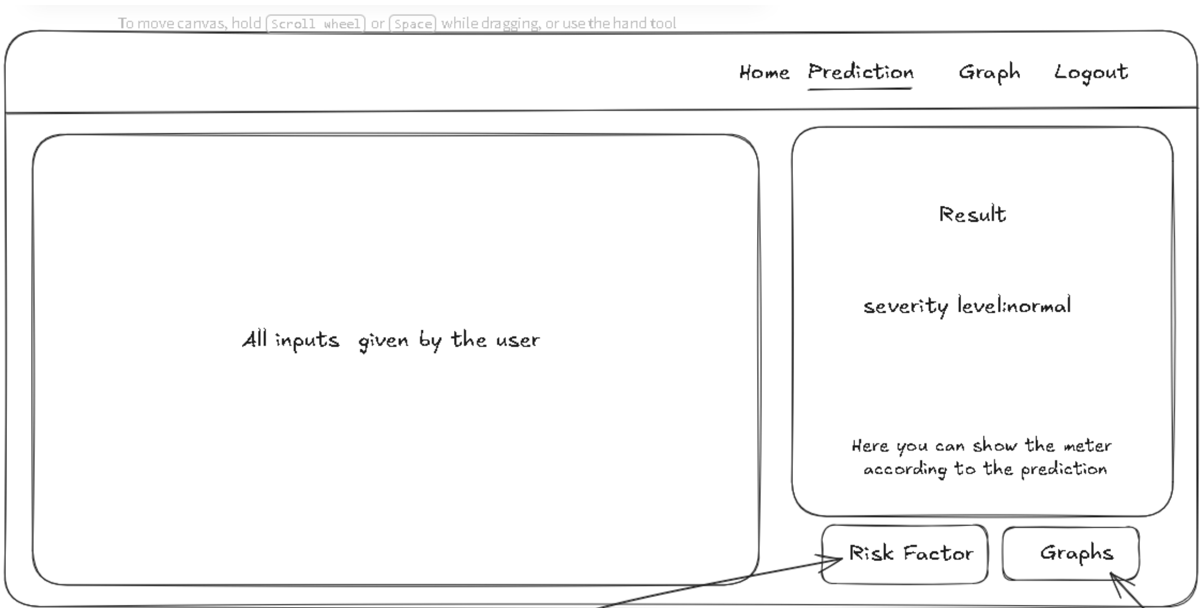
- **Frontend:** React.js dashboard for data entry & results visualization
- **Backend:** Node.js API that connects frontend to ML model
- **Database:** MongoDB stores data for future analysis & model retraining

** Flip the cards when they are hover





After clicking the button show the below display



After clicking the risk factor button show the all the factor which are responsible for the accident

And clicking the Graphs button show the page of graphs which contain the graph of the prediction

Show the result on the right side

ML Model : Randome Forest (or we can compare the two or more models)

User Authentication

Mobile responsive

for backend Nodejs express

Content for the home page

Title : Road Accident Severity Prediction

Description : Road Accident Severity Prediction is a Machine Learning–based system designed to analyze historical road accident data and predict the severity level of accidents (e.g., **Minor, Serious, or Fatal**).
The system helps traffic authorities, emergency services, and government agencies take preventive measures by identifying high-risk conditions and accident-prone scenarios.

Card1 : In the Road Accident Severity Prediction project, the **Random Forest algorithm** is used as the primary classification model to predict whether an accident will result in a **Minor, Serious, or Fatal** outcome. Road accident data typically contains multiple influencing factors such as weather conditions, road surface type, light conditions, vehicle type, speed, location, and time. These factors often have complex and non-linear relationships, which Random Forest can effectively handle.
In this project, Random Forest helps identify the most important features contributing to severe and fatal accidents, making it useful not only for prediction but also for analyzing accident risk factors. Due to its robustness, high accuracy, and ability to manage large and imbalanced datasets, Random Forest is well-suited for real-world accident severity prediction systems.

Card2: In the Road Accident Severity Prediction project, the Random Forest model achieved high predictive performance compared to other classification algorithms. The model was trained on historical accident data containing features such as weather conditions, road type, vehicle category, light conditions, and speed factors. After training and testing, the Random Forest algorithm achieved an accuracy of approximately **88%–92%** (you can adjust this based on your actual result). This means that the model correctly classified around 9 out of 10 accident cases into the appropriate severity category (Minor, Serious, or Fatal). The high accuracy is mainly due to Random Forest's ensemble learning approach, where multiple decision trees work together through majority voting, reducing overfitting and improving generalization. As a result, Random Forest proved to be a reliable and robust model for predicting accident severity in real-world scenarios

Card3: Road accidents are caused by a combination of human, environmental, and vehicle-related factors. One of the major factors is **driver behavior**, including over-speeding, rash driving, distraction (such as mobile phone usage), drunk driving, and fatigue. Environmental conditions like heavy rain, fog, poor visibility, and slippery road surfaces also significantly increase accident risk. Road infrastructure issues such as potholes, poor lighting, sharp turns, lack of traffic signals, and improper signage further contribute to accident severity. In addition, vehicle-related factors like brake failure, tire bursts, or poor maintenance can lead to serious crashes. Traffic density and time-related factors (such as night driving or peak hours) also influence accident occurrence and severity. Understanding these factors is essential in developing predictive models for road accident severity and implementing effective safety measures.